

Lab 5. Configuring File and Folder Permissions. Sharing Folders.

IT Infrastructure Management (W04IST–SI0826G)

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Spring 2026

1 Lab Objectives

1. Preparing a corporate environment project.

2 Introduction

As part of this task, each group prepares a configuration for a company of their choice.

3 Exercises

In this task you will learn how to configure permissions for folders and files stored on an NTFS partition.

3.1 Configuring NTFS Permissions

1. Log in to computer CL1 with an administrator account.
2. Create the folder `c:\localdata`. Open the folder properties, go to the “Security” tab and click “Advanced”.
3. Review the Access Control List (ACL). See who has access to the folder. Where are the individual entries inherited from?
4. Disable permission inheritance using the “Disable Inheritance” button, converting the inherited permissions into permissions assigned directly to the folder (“Convert...”). Remove all entries except: SYSTEM, Administrators, and Users:: Read and execute.
5. Create the folder `c:\localdata\forall`.
6. Create the folder `c:\localdata\forSwedenMarketing`. Disable permission inheritance by copying the existing entries. Remove the Users entry. Add an entry for the group “G_Sweden_Marketing”:: Allow:: Full control and “U_Marketing”:: Allow:: Read.
7. Log in with user accounts from the following OUs: Sweden/Marketing, Sweden/Sales, Norway/Marketing and check what you can do in each folder.

3.2 Simple Folder Sharing

In this exercise you will share a folder on the network using simple folder sharing. A folder shared this way has: the same permissions set in all subfolders and a name identical to the folder on disk.

1. Log in to computer DC1 with an administrator account.
2. Create the folder `c:\reports-simple` and share it using **simple folder sharing** for the group “U_Marketing” with Read permission and for the group “G_Sweden_Marketing” with Read/Write permission.

Hint: Folder context menu > Give access to > Specific people.

3. Log in to machine CL1 with accounts from the OUs Sweden/Marketing and Norway/Marketing. Connect to the reports-simple folder shared on server DC1 and check what you can do in that folder. Connect using a different method each time:
 - Open the shared folder directly in Windows Explorer: type the network folder path in the address bar: `\\dc1\reports-simple`,
 - Map the shared folder as a new drive S: using Windows Explorer: click the This PC icon, then expand the Computer ribbon and click Map Network Drive,
 - Map the shared folder as a new drive T: using the command line (CMD):

```
c:\> net use T: \\dc1\reports-simple
```
4. Verify that drives S: and T: are visible in Windows Explorer.
5. Check whether you can create a new file or edit an existing file on drives S: and T:.
6. Disconnect drive S: by selecting Disconnect from the context menu.
7. Disconnect drive T: using the command `net use /d T:`

3.3 Advanced Folder Sharing

Advanced sharing is used when you want to set different permissions for subfolders or share a folder under a different name than it has on disk. Advanced sharing is configured by setting two sets of permissions:

1. Share permission — a common permission for the entire share (similar to simple sharing),
2. NTFS permission — permissions for specific folders or files on disk (as in the first exercise).

The effective permission is the intersection of both sets of permissions.

1. Log in to computer DC1 with an administrator account.
2. Create the folder `c:\reports-advanced` and share it with users using **advanced folder sharing**. Add the permission for the group: Everyone:: Full Control.

Hint: Folder context menu > Properties > Share tab > Advanced Sharing button.

3. Following exercise 1, set NTFS permissions for the folder **C:\reports-advanced** as follows:
 1. Disable permission inheritance from drive **C:**, keeping the existing permissions.
 2. Remove all permissions except SYSTEM and Administrators.
 3. Add the permission “U_Marketing”:: Allow:: Read & execute.
 4. Add subfolders: **2022**, **2023**, ..., **current**.
 5. In the **current** folder, add a new permission “G_Sweden_Marketing”:: Allow:: Full control.
 6. Create one text file in each folder.
4. Connect to the shared folders from machine CL1 using accounts from the Sweden/Marketing and Norway/Marketing organizational units.

Check whether you can enter the **current** folder and edit the file. Do the same for the **2022** folder.

Final note: **Simple sharing** (as in exercise 2) automatically sets the “Share” permissions to Everyone:: Allow:: Full control, and the NTFS permissions according to the settings in the Give access to > Specific people window. Every time permissions are modified, all NTFS permissions in subfolders are reset.

With **advanced sharing**, the “Share” and “NTFS” permissions can be modified independently of each other.

3.4 Sharing Folders via PowerShell

In this exercise you will see how to share a folder and connect to it using PowerShell.

1. On computer DC1, use the PowerShell console to create a new folder and share it on the network:

```
PS1:\>New-Item C:\reports-ps -Type Directory
PS1:\>New-SMBShare -Name "reports-powershell" -Path c:\reports-ps
↵ -FullAccess "G_Sweden_Marketing" -ReadAccess "U_Marketing"
```

2. On computer CL1, log in with accounts from the OUs Sweden/Marketing, Norway/Marketing, Norway/Sales and map the shared folder as a drive:

```
PS1:\>New-PSDrive -Name R -PSProvider FileSystem -Root
↵ "\\dc1\reports-powershell" -Persist
```

Note: Without the **-Persist** option, the new drive will only be visible in the current PowerShell session.

3. Check:
 - whether you can see drive R: in Windows Explorer,
 - what operations you can perform on drive R:.